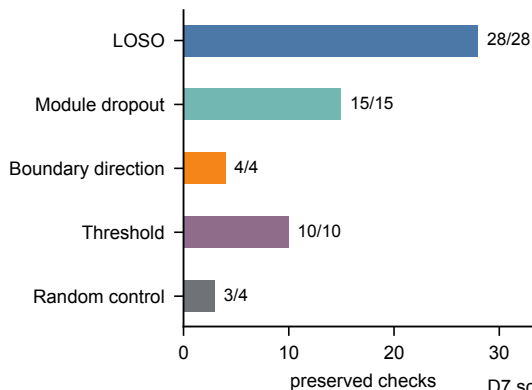


b

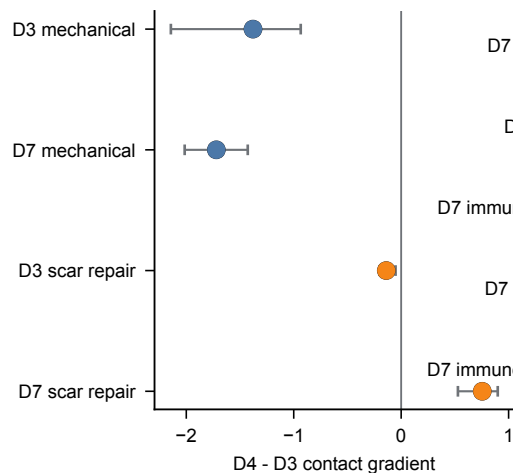
a

Public-data robustness

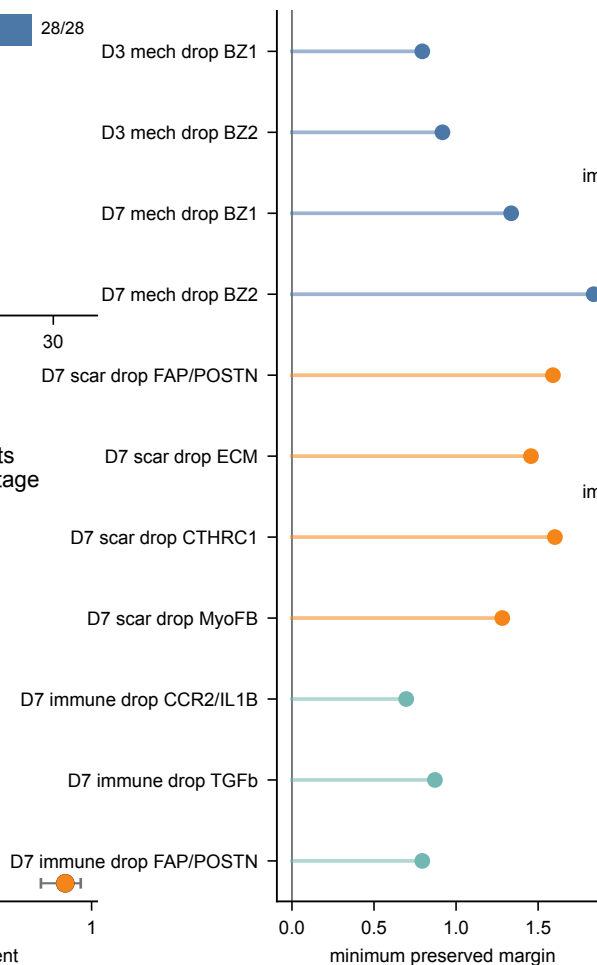


c

Boundary gradients change with repair stage

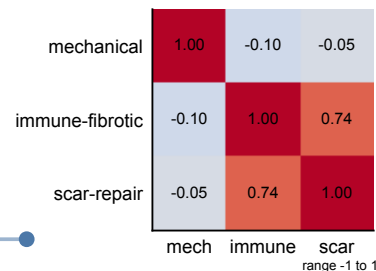


No single module drives the axis call

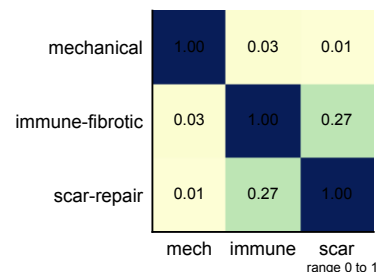


d

Spearman r



Top-decile Jaccard



Human STEMI transfer summary

Mechanical-border and fibroblast-scar axes exceeded random controls. The immune-fibrotic axis was directionally positive but treated as a contextual coupling signal. Human STEMI maps showed distinct mechanical hotspots and partial immune-scar overlap, supporting three outputs.